AI-Based Autonomous Space Monitoring System

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Outline

1. Introduction
2. Methodology
3. Results
4. Conclusion

Introduction

- Deep space missions experience communication delays.
- Real-time decisions cannot rely on Earth.
- AI enables autonomous monitoring and control.

Problem Statement

- Detect anomalies in spacecraft telemetry.
- Predict system failure in advance.
- Recommend corrective actions autonomously.

Objectives

- Anomaly Detection
- Remaining Useful Life Prediction
- Risk Scoring System
- Autonomous Decision Engine

Dataset

- NASA CMAPSS Turbofan Dataset
- Multi-sensor time-series data
- Simulates engine degradation

System Architecture

Pipeline Steps

1. Sensor Data Collection
2. Feature Engineering
3. RUL Prediction Model
4. Anomaly Detection
5. Risk Scoring
6. Decision Engine

Feature Engineering

- Rolling Mean and Standard Deviation
- Lag Features for time dependency
- Trend-based degradation features

RUL Prediction Model

- XGBoost Regressor
- Captures nonlinear degradation
- Achieved $R^2 \approx 0.80$

Anomaly Detection

- Isolation Forest (unsupervised)
- Hybrid anomaly score
- Balanced precision and recall

Risk Scoring

- Combines anomaly score, RUL, and sensor instability
- Provides continuous risk estimation
- Used for decision making

Decision Engine

- Risk > 0.8 - Critical
- Risk > 0.6 - Warning
- Else - Normal

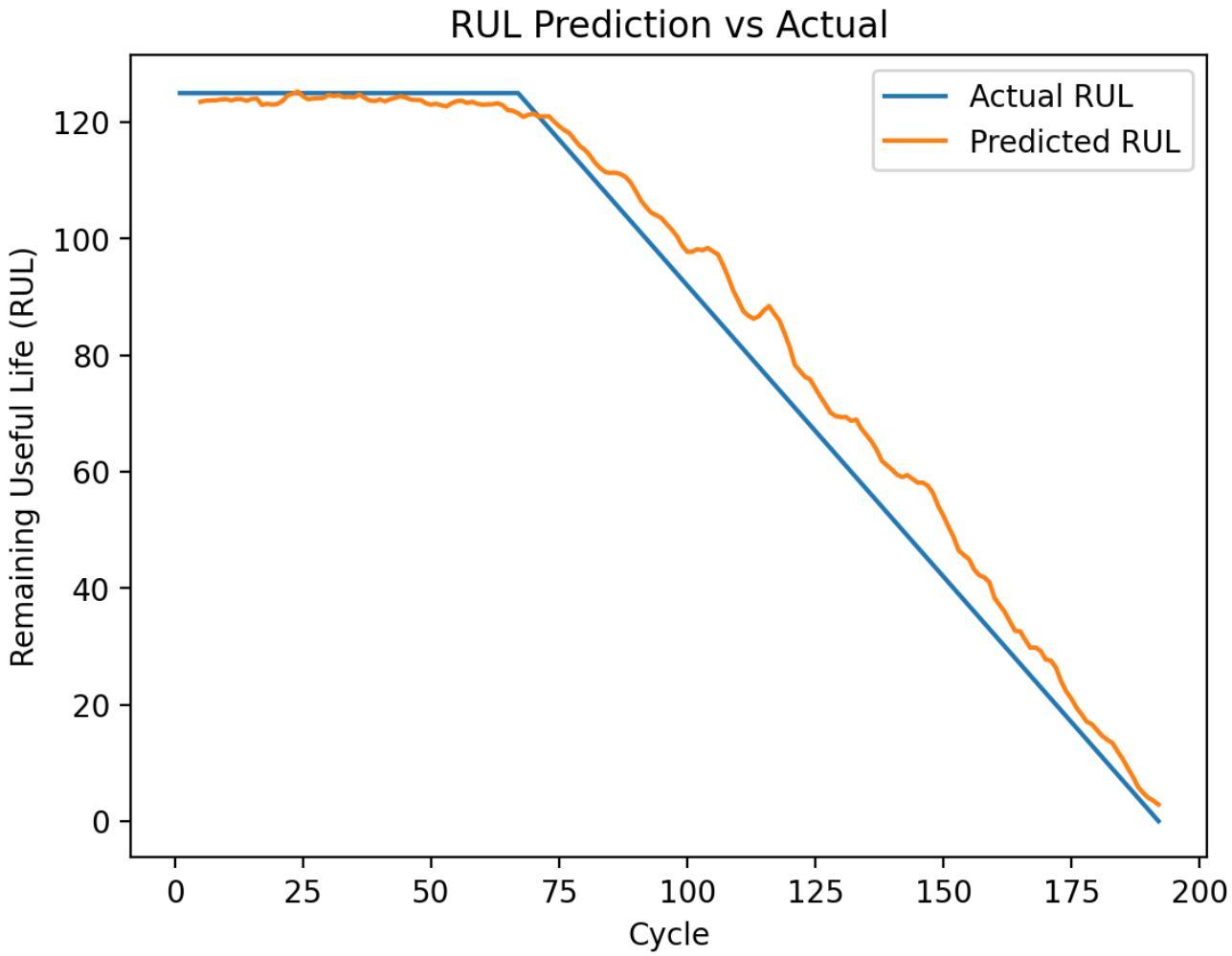
Automatically triggers system-level responses based on risk.

Model Performance

- RMSE ≈ 18
- MAE ≈ 13
- $R^2 \approx 0.80$
- Anomaly F1-score ≈ 0.82

Automatically triggers system-level responses based on risk.

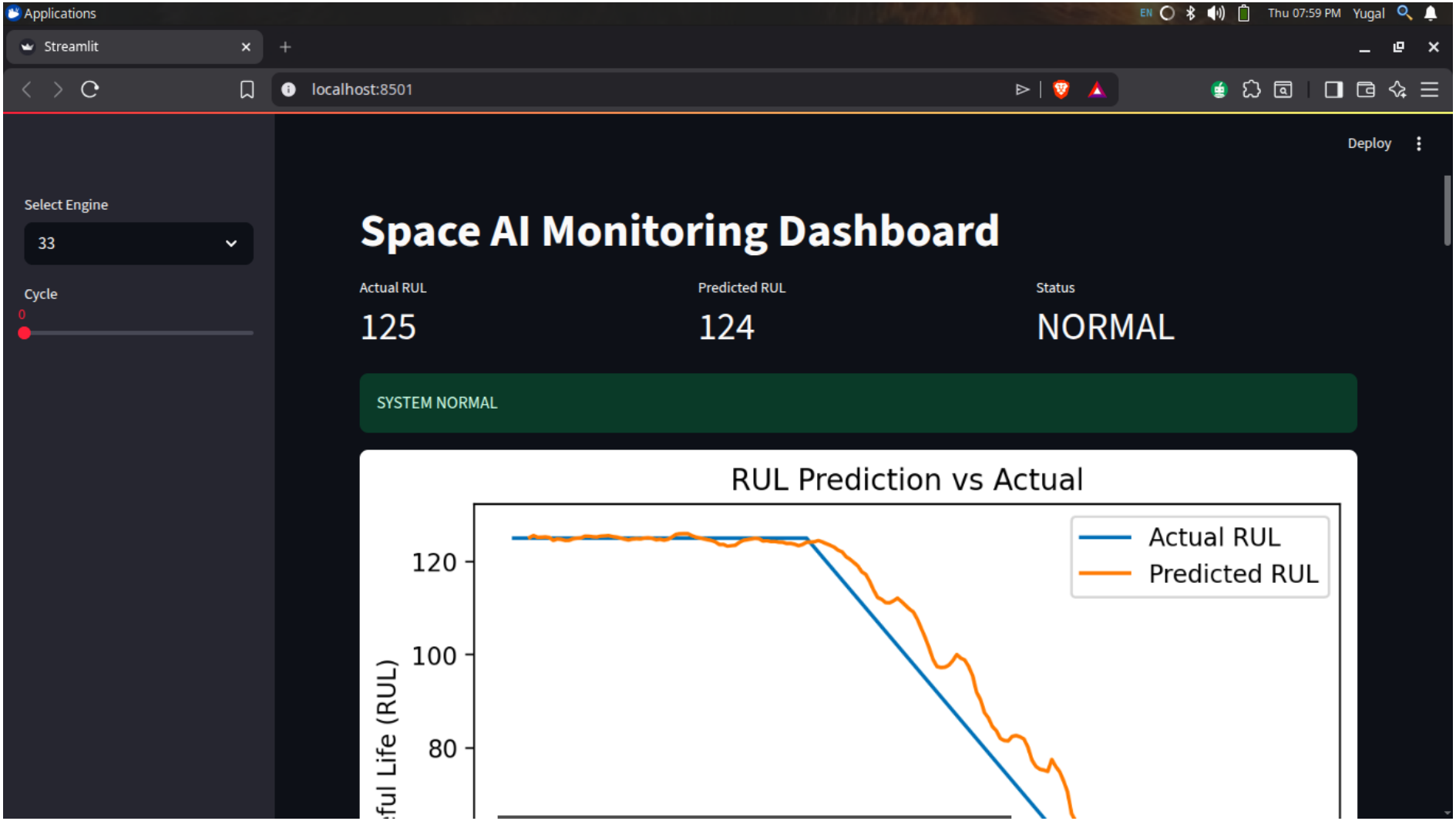
RUL Prediction Graph



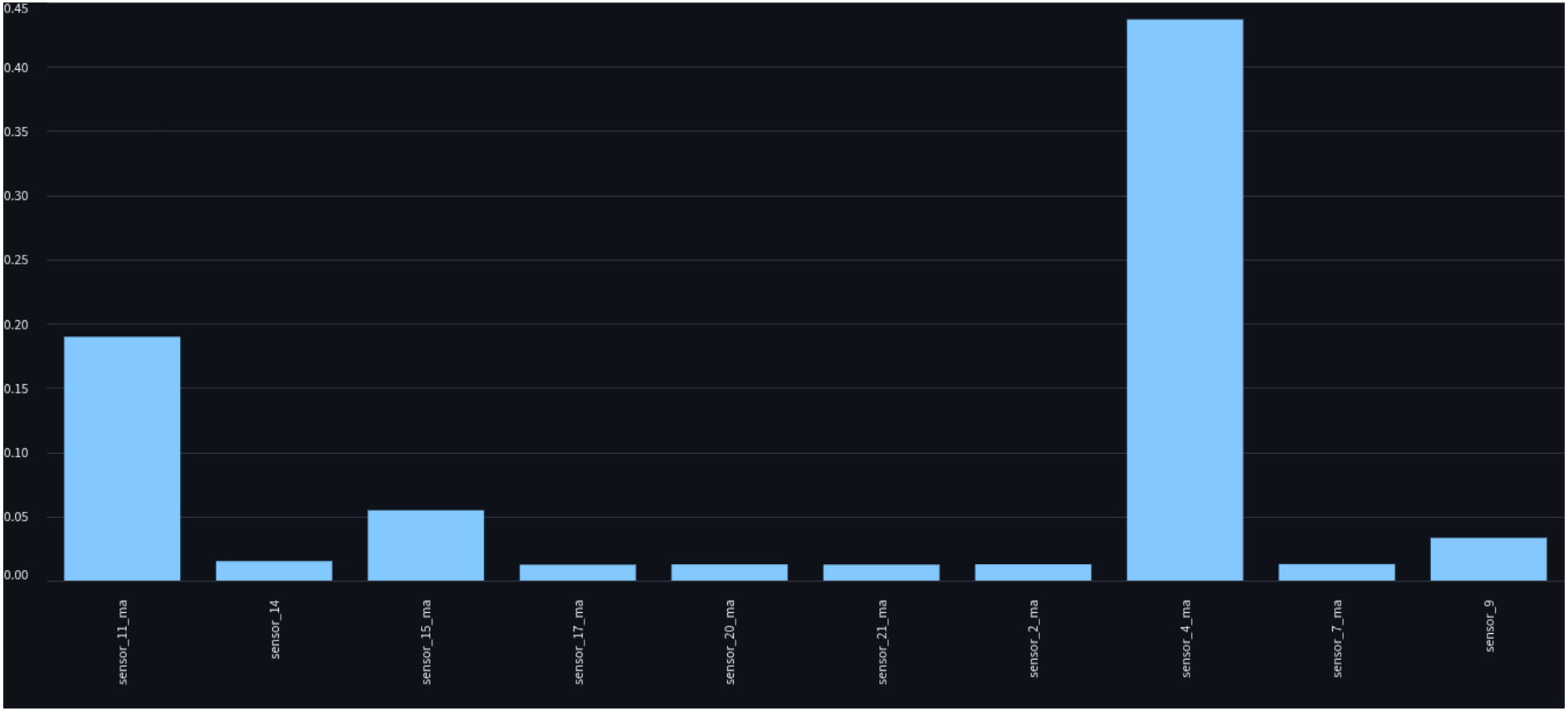
Dashboard Overview

- Interactive monitoring of engine health
- Visualization of RUL, risk, and anomalies
- Supports real-time decision-making

Dashboard Screenshot



Top Features for Anomaly Detection



Anomaly Detection Performance

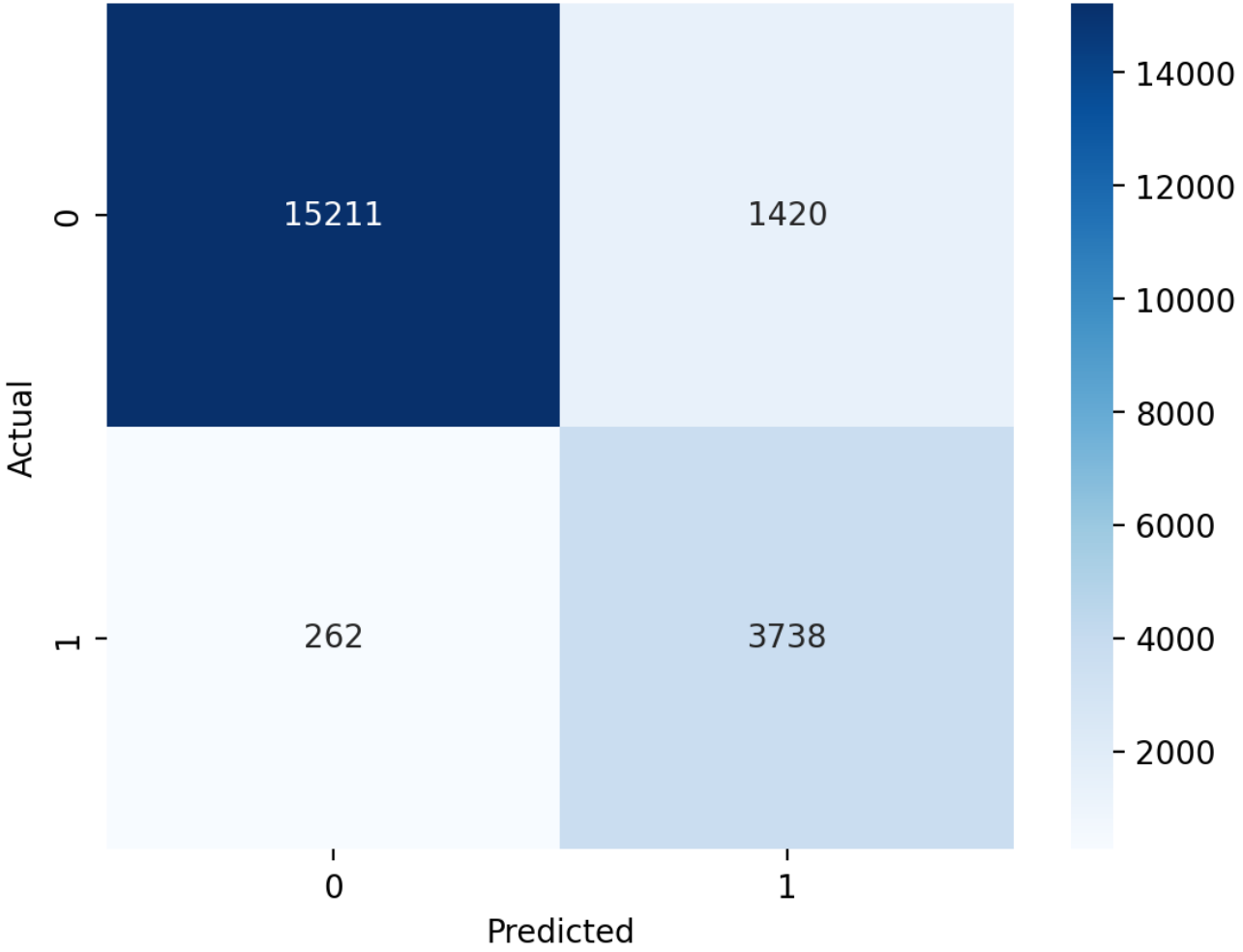
- Accuracy: 0.92
- Precision: 0.72
- Recall: 0.93
- F1 Score: 0.82

High recall ensures most anomalies are detected.

Interpretation

The model achieves a balance between precision and recall, ensuring reliable anomaly detection while minimizing missed failures.

Confusion Matrix



Conclusion

- High recall ensures most anomalies are detected.
- Effectively detects anomalies in telemetry data
- Enables autonomous decision support

Suitable for real-time space mission applications

Future Work

- LSTM-based time series models
- Autoencoder anomaly detection
- Real-time telemetry integration